DATA WAREHOUSING AND MINING LABORATORY

Experiment No. 2: Create and Analyze ARFF Files Using WEKA

Aim: To create, load, and analyze ARFF (Attribute Relation File Format) datasets using WEKA and understand the structure and representation of data used in data mining applications.

Objectives: After successful completion of this experiment, students will be able to:

1. Understand the concept of dataset representation in data mining.
2. Define ARFF and its importance in WEKA.
3. Understand the structure of an ARFF file.
4. Differentiate between relations, attributes, and instances.
5. Create ARFF files manually using a text editor.
6. Define numeric and nominal attributes.
7. Load ARFF files into WEKA.
8. Analyze the contents of a dataset.
9. Identify and correct common ARFF syntax errors.

Expected Outcome: After completing this experiment, Students will be able to:

1. Create their own datasets.
2. Save datasets in ARFF format.
3. Load custom datasets into WEKA.
4. Analyze dataset properties.
5. Prepare datasets required for preprocessing, association mining, classification, and clustering experiments.

Importance of this Experiment:

Data mining algorithms work only when data is represented in a proper format. Before performing Data Cleaning, Association Rule Mining, Classification, Clustering students must understand how datasets are structured and stored. This experiment provides the foundation for all future DWDM laboratory experiments.

Facilities Required:

Hardware Requirements:

- Personal Computer / Laptop
- Intel i3 Processor or higher
- Minimum 4 GB RAM
- Internet Connection (Optional)

Software Requirements:

- Windows 10/11
- WEKA Tool
- Notepad / Notepad++

Prerequisite Knowledge: Students should be familiar with:

- Experiment No. 1
- Basic WEKA environment
- Attributes and Instances
- Data Mining Fundamentals

Theory:

Dataset Representation in Data Mining:

Data Mining algorithms cannot process raw information directly. Before analysis, data must be organized into a structured dataset. Common dataset formats include:

- CSV (Comma Separated Values)
- Excel Files
- ARFF Files

WEKA primarily uses ARFF format.

What is ARFF?

ARFF stands for Attribute Relation File Format. ARFF is the standard file format used by WEKA to store datasets. It contains:

- Relation Name
- Attribute Definitions
- Dataset Records

ARFF files help WEKA understand:

- Dataset Structure

- Data Types
- Class Labels
- Attribute Information

Structure of an ARFF File:

Every ARFF file consists of two major sections:

1. Header Section

Contains:

- Relation Name
- Attribute Definitions

2. Data Section

Contains:

- Dataset Records (Instances)

General Structure:

@relation relation_name

@attribute attribute_name datatype

@data

value1,value2,value3

value1,value2,value3

Components of an ARFF File:

Relation: Represents the dataset name.

Example: @relation student

The relation name should describe the dataset.

Attribute: Represents a characteristic or feature of data.

Example: @attribute attendance numeric

Here: attendance is the attribute name.

Numeric Attribute: Stores numerical values.

Examples: attendance, marks, studyhours, salary

Declaration: @attribute attendance numeric

Nominal Attribute: Stores categorical values.

Example: @attribute result {Pass,Fail}

Possible values: Pass, Fail

Data Section: Contains actual dataset records.

Example: @data

80,5,Pass

60,2,Fail

90,6,Pass

Each row is called an Instance.

Industry Insight: Before building any Machine Learning or Data Mining model, data must be collected, structured, validated, and represented correctly. In real-world projects, data preparation often consumes more effort than model development.

Mini Real-World Example:

Student Performance Prediction: Suppose we want to predict whether a student will pass or fail.

Dataset Attributes: Attendance, Study Hours, Result

Dataset: Attendance, StudyHours, Result

80,5,Pass

60,2,Fail

90,6,Pass

70,3,Pass

50,1,Fail

Before applying classification algorithms, the dataset must be represented correctly in ARFF format.

Sample ARFF Dataset:

Student Performance Dataset

@relation student

@attribute attendance numeric

@attribute studyhours numeric

@attribute result {Pass,Fail}

@data

80,5,Pass

60,2,Fail

90,6,Pass

70,3,Pass

50,1,Fail

Procedure:

Activity 1: Creating ARFF File Using Notepad

Step 1: Open Notepad.

Step 2: Type the following ARFF dataset:

@relation student

@attribute attendance numeric

@attribute studyhours numeric

@attribute result {Pass,Fail}

@data

80,5,Pass

60,2,Fail

90,6,Pass

70,3,Pass

50,1,Fail

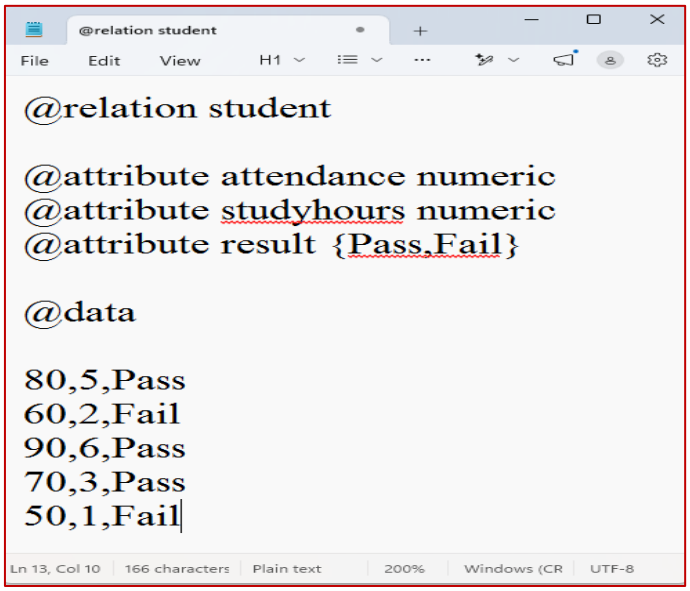


Figure 2.1: Creating ARFF File in Notepad

Step 3: Select: File → Save As

Step 4: Enter filename: student.arff

Step 5: Select: Save as Type = All Files

Step 6: Click Save.

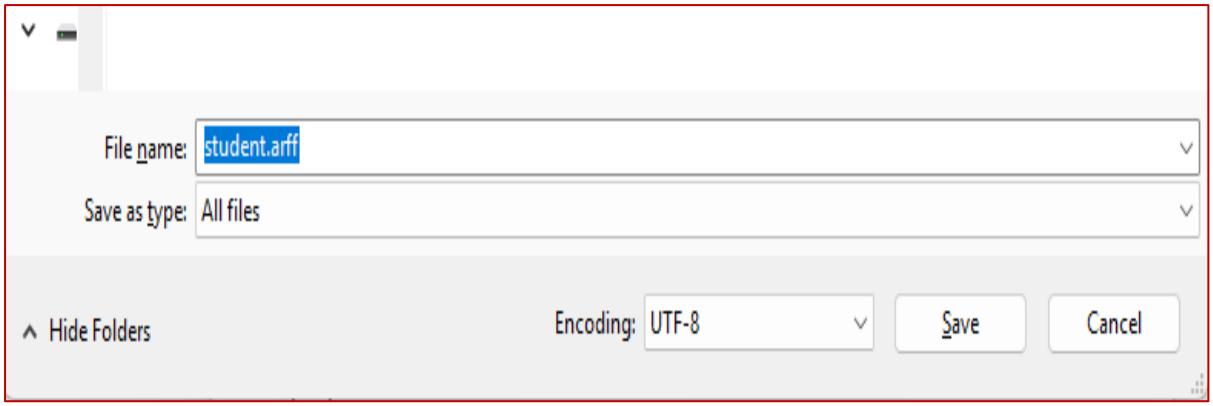


Figure 2.2: Saving student.arff File

Activity 2: Loading ARFF File into WEKA

Step 1: Open WEKA.

Step 2: Click: Explorer

Step 3: Select: Preprocess Tab

Step 4: Click: Open File

Step 5: Select: student.arff

Step 6: Click Open.

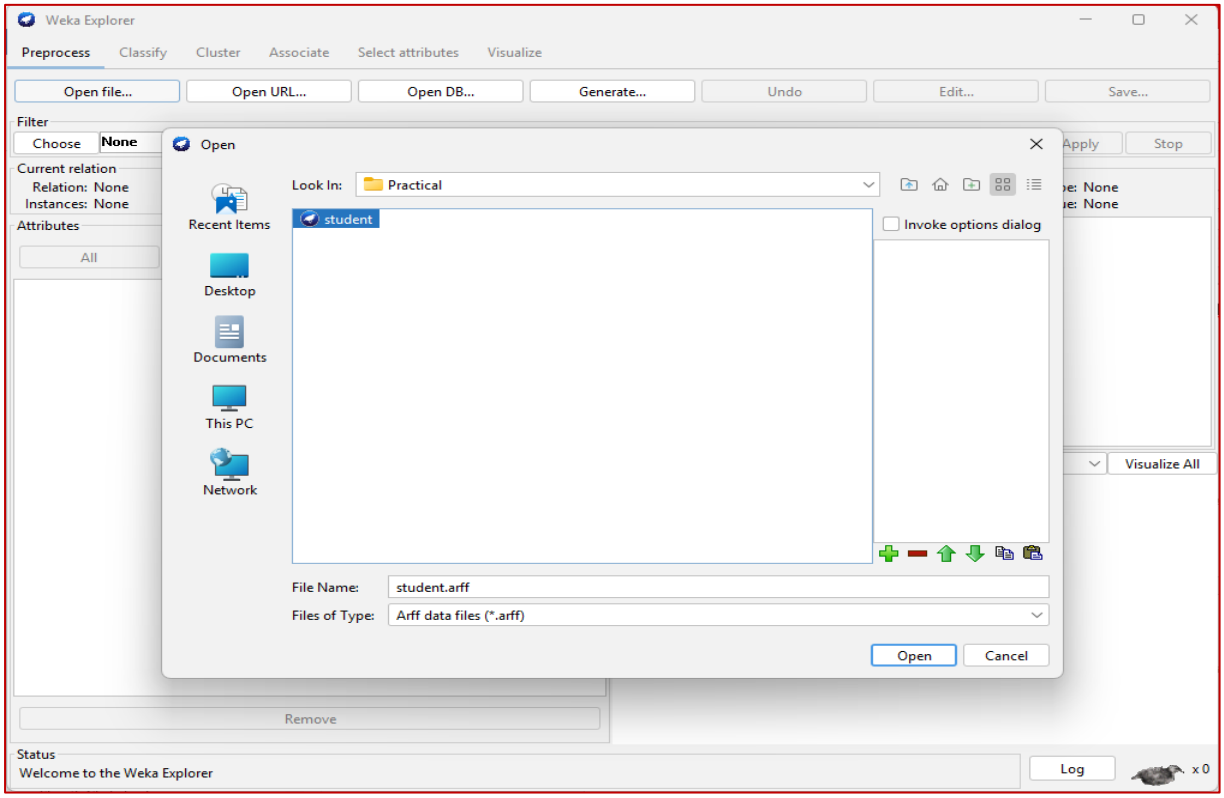


Figure 2.3: Loading ARFF File into WEKA

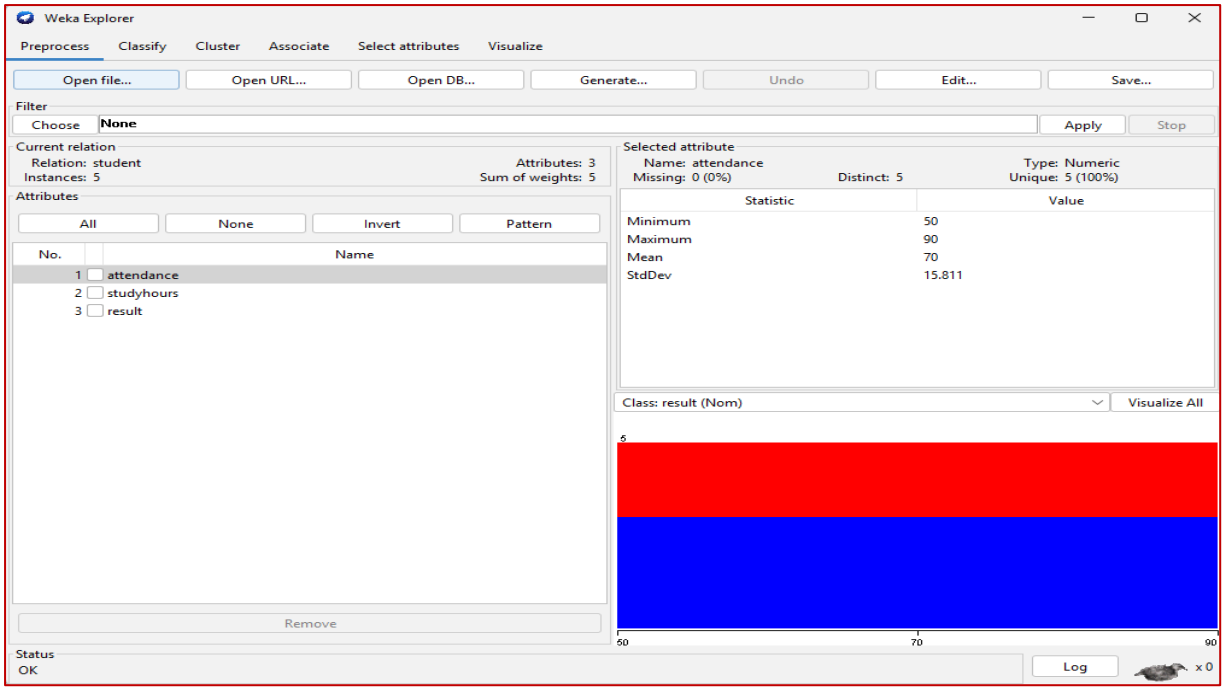


Figure 2.4: Dataset Loaded Successfully

Activity 3: Analyzing Dataset

Observe the following:

Relation Name: student

Number of Attributes: 3

Attributes

- attendance
- studyhours
- result

Number of Instances: 5

Class Attribute: result

A Class Attribute is the target attribute that WEKA attempts to predict during classification. In this experiment, result is selected as the class attribute because its values (Pass or Fail) can be predicted using attendance and studyhours in future classification experiments.

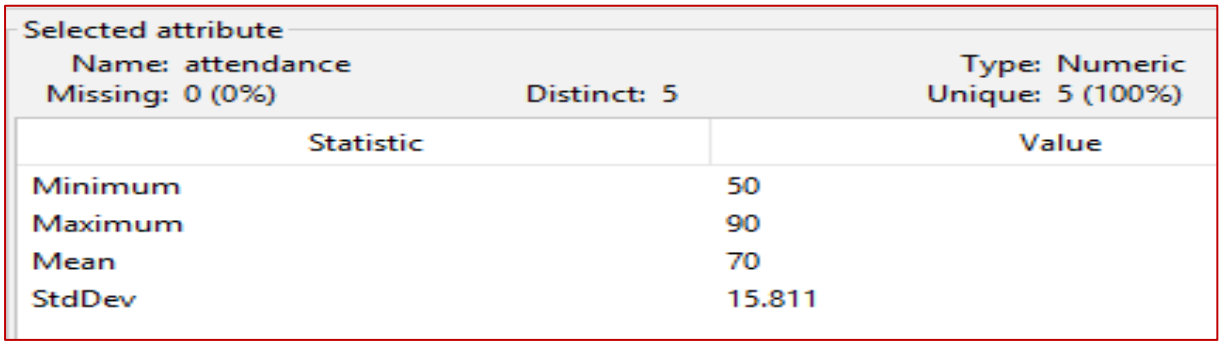


Figure 2.5: Statistical Information of Attribute "attendance"

WEKA displays statistical details of the selected attribute such as Minimum, Maximum, Mean, and Standard Deviation, which help in understanding the distribution of values in the dataset.

Common ARFF Errors:

- Error 1: Missing Braces

  Wrong: @attribute result Pass Fail

  Correct: @attribute result {Pass,Fail}

- Error 2: Missing Commas

  Wrong: 80 5 Pass

  Correct: 80,5,Pass

- Error 3: Invalid Data Type

  Wrong: @attribute attendance number

  Correct: @attribute attendance numeric

- Error 4: Missing @data Section

  Wrong:

  @relation student

  @attribute attendance numeric

  80,5,Pass

  Correct:

  @relation student

  @attribute attendance numeric

  @data

  80,5,Pass

Student Activity:

Create any one of the following datasets in ARFF format:

Option A: Student Performance Dataset

Option B: Library Management Dataset

Option C: Mobile Purchase Dataset

Requirements:

- Minimum 3 Attributes

- Minimum 5 Instances
- At least one Numeric Attribute
- At least one Nominal Attribute

Load the dataset successfully in WEKA and verify all attributes.

Observation Table:

PARAMETER	OBSERVATION
Relation Name	
Number of Attributes	
Number of Instances	
Numeric Attributes	
Nominal Attributes	
Class Attribute	
Dataset File Name	

Result: An ARFF file was successfully created, loaded, and analyzed using WEKA. The structure of ARFF files, including relations, attributes, and instances, was studied successfully.

Viva Questions:

1. What is ARFF?
2. Expand ARFF.
3. Why is ARFF used in WEKA?
4. What is a relation?
5. What is an attribute?
6. What is an instance?
7. Differentiate between numeric and nominal attributes.
8. What is the purpose of the @data section?
9. What happens if @data is omitted?
10. What is a class attribute?

11. How are nominal values represented in ARFF?
12. Can WEKA load CSV files?
13. What is the purpose of the header section?
14. What is the purpose of the data section?
15. Why is proper dataset representation important?

Assignment Questions:

1. Create an ARFF file for employee information.
2. Create an ARFF file for library management data.
3. Explain the structure of an ARFF file with a suitable example.
4. Differentiate between CSV and ARFF.
5. Create an ARFF dataset containing five attributes and ten instances.
6. Design an ARFF dataset that can be used for placement prediction.

References:

1. Ian H. Witten, Eibe Frank and Mark A. Hall, *Data Mining: Practical Machine Learning Tools and Techniques*, Morgan Kaufmann.
2. Jiawei Han, Micheline Kamber and Jian Pei, *Data Mining: Concepts and Techniques*, Morgan Kaufmann.
3. WEKA Official Documentation & User Manual.

Prof. F I Khandwani	**Dr. S D Padiya**
Subject In-charge	**Head of the Department**